CBSE Class 10 Mathematics — Arithmetic Progressions

Time Allowed: 3 Hours **Maximum Marks: 80**

General Instructions:

1. This question paper contains $_{questions. All questions are compulsory.}$

1. Questions are grouped into sections A through E.

2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. The 4th term of an AP is 14 and the 12th term is 70. What is the first term a?

(A) (A) -7
(B) (B) 7
(C) (C) -14
(D) (D) 14

2. If the sum of first n terms of an AP is $S_n = 3n^2 + n$, what is the common difference d?

(A) (A) 3
(B) (B) 4
(C) (C) 6
(D) (D) 2

3. Which term of the AP: 21, 18, 15, ... is -81?

(A) (A) 34th
(B) (B) 35th
(C) (C) 36th
(D) (D) 37th

4. In an AP, if $a = 5$ and $d = 3$, find the sum of the first 10 terms (S_{10})?

(A) (A) 150
(B) (B) 175
(C) (C) 185
(D) (D) 195

5. If $k, 2k - 1, 2k + 1$ are three consecutive terms of an AP, find the value of k.

(A) (A) 3
(B) (B) 2
(C) (C) 4
(D) (D) 5

6. Find the sum of all odd numbers between 0 and 50.

(A) (A) 600
(B) (B) 625
(C) (C) 650
(D) (D) 675

7. The 10th term from the end of the AP: 5, 8, 11, ..., 185 is:

(A) (A) 150
(B) (B) 152
(C) (C) 155
(D) (D) 158

8. If the sum of first n terms is n^2, then the nth term is:

(A) (A) 2n+1
(B) (B) 2n
(C) (C) 2n-1
(D) (D) n-1

9. How many two-digit numbers are divisible by 3?

(A) (A) 30
(B) (B) 31
(C) (C) 32
(D) (D) 33

10. What is the common difference of an AP in which $a_{25} - a_{20} = 45$?

(A) (A) 5
(B) (B) 9
(C) (C) 10
(D) (D) 4.5

11. Assertion (A): The common difference of the AP $5, 2, -1, -4, \ldots$ is -3. Reason (R): The common difference d of an AP $a_1, a_2, a_3, \ldots$ is given by $d = a_n - a_{n-1}$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

12. Assertion (A): If the n^{th} term of an Arithmetic Progression is given by $a_n = 3 + 4n$, then the common difference of the AP is 4. Reason (R): The common difference d of an AP is given by the difference between consecutive terms $a_n - a_{n-1}$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

13. Assertion (A): The sum of the first n terms of an arithmetic progression is given by $S_n = \frac{n}{2}[2a+(n-1)d]$, where a is the first term and d is the common difference. Reason (R): The sum of the first n terms of an AP can be calculated as $S_n = \frac{n}{2}(\text{first term} + \text{last term})$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

14. Assertion (A): The sum of the first n terms of an arithmetic progression is given by $S_n = 3n^2 + 5n$. The common difference of this AP is 6. Reason (R): For an AP where $S_n = an^2 + bn$, the common difference is always $2a$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

15. Assertion (A): The 10th term of the AP: $5, 8, 11, 14, \ldots$ is 32. Reason (R): The nth term of an AP with first term a and common difference d is given by $a_n = a+(n-1)d$.

(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)

(B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

(C) (C) Assertion (A) is true but Reason (R) is false

(D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. Find the 11^{th} term of an Arithmetic Progression (AP) whose first term is -3 and the common difference is 4.

2. Which term of the AP $21, 18, 15, \ldots$ is -81?

3. Find the sum of the first 10 multiples of 5.

4. If the n^{th} term of an AP is given by $a_n = 3 + 2n$, find its common difference.

5. Find the value of k for which the terms $k-1, k+3, 3k-1$ are in Arithmetic Progression.

Section D: Long Answer (4-5 marks each)

1. Find the sum of all two-digit numbers which, when divided by 4, leave 1 as a remainder.

2. If the sum of the first n terms of an AP is $S_n = 3n^2 + 5n$, find the 16th term of the AP.

3. The 4th term of an AP is 0. Prove that the 25th term of the AP is three times its 11th term.

4. How many terms of the AP 24, 21, 18, ... must be taken so that their sum is 78?

5. Find the middle term of the Arithmetic Progression 7, 13, 19, ..., 241.

6. The sum of the first n terms of an Arithmetic Progression is given by $S_n = 3n^2 + 5n$. Find the n^{th} term of this AP and hence find the 20^{th} term.

7. In an AP, the sum of first 10 terms is -150 and the sum of its next 10 terms is -550. Find the Arithmetic Progression.

8. $Asumof$Rs. $1600istobeusedtogive5cashprizestostudentsofaschoolfortheiroverallacademicperfor$

9. Find the sum of all three-digit numbers which leave a remainder of 3 when divided by 5.